

$$X \sim \text{SubG}(\sigma) \quad \mathbb{E}(X) = 0$$

$$Z \sim \mathcal{N}(0, 2\sigma^2)$$

$$\exists c \quad \underbrace{\mathbb{P}(X \geq t)}_{\substack{\text{tail prob.} \\ X}} \leq c \underbrace{\mathbb{P}(Z \geq t)}_{\substack{\text{tail prob.} \\ Z}}$$

Lemma. Mill's Ratio

$$Z \sim \mathcal{N}(0, 1) \quad \mathbb{P}(Z \geq t) \geq \left(\frac{1}{t} - \frac{1}{t^3}\right) \phi(t)$$

$$\phi(\cdot) \text{ density of } Z = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

$$\text{Proof: } \frac{\mathbb{P}(X \geq t)}{\underbrace{\mathbb{P}(Z \geq t)}_r} \leq c.$$

$$r \leq \frac{\exp\left\{-\frac{t^2}{2\sigma^2}\right\}}{\frac{1}{\sqrt{2\pi}} \left(\frac{\sqrt{2}\sigma}{t} - \frac{2\sqrt{2}\sigma^3}{t^3}\right) e^{-t^2/4\sigma^2}}$$

$$Z \sim \mathcal{N}(0, 2\sigma^2)$$

$$\mathbb{P}\left(\frac{Z}{\sqrt{2}\sigma} \geq \frac{t}{\sqrt{2}\sigma}\right)$$

$$\underbrace{Z'}_{Z' \sim \mathcal{N}(0, 1)}$$

$$\stackrel{(\text{Mills})}{\geq} \left(\frac{\sqrt{2}\sigma}{t} - \frac{2\sqrt{2}\sigma^3}{t^3}\right)$$

$$\frac{1}{\sqrt{2\pi}} e^{-t^2/4\sigma^2}$$

Case 1: $0 < t < 2\sigma$

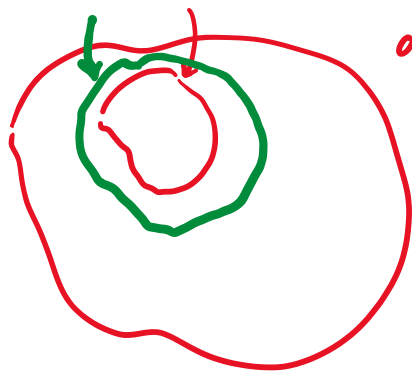
$$r \leq \frac{1}{\frac{1}{\sqrt{2\pi}} \left(\frac{\sqrt{2}}{2} - \frac{2\sqrt{2}}{8}\right) e^{-1}}$$

$$= \frac{\sqrt{2\pi} e \frac{4}{\sqrt{2}}}{c} = \frac{4\sqrt{\pi} e}{c}$$

Case 2 : $t > 2\sigma$

$$r \leq \frac{\sqrt{2\pi} e^{-t^2/4\sigma^2}}{\left(\frac{\sqrt{2}\sigma}{t} - \frac{2\sqrt{2}}{t^3}\sigma^3\right)} \leq 4\sqrt{\pi} e.$$

SubE SubG



all dist

$\forall \lambda \quad \mathbb{E}(e^{\lambda(X-\mu)})$ exist

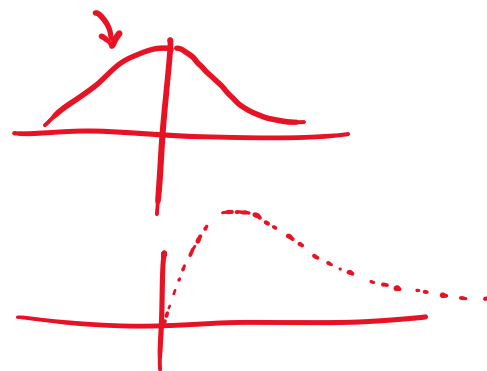
Def. $X \sim \text{SubE}(\nu, \alpha)$

$$\mathbb{E}(e^{\lambda(X-\mu)}) \leq e^{\nu^2 \lambda^2 / 2} \quad |\lambda| \leq \frac{1}{\alpha}$$

Example :

$$Z \sim \mathcal{N}(0, 1)$$

$$Z^2 \sim \chi$$



$$\mathbb{E}(e^{\lambda(Z^2-1)})$$

$$= \int_{-\infty}^{+\infty} e^{\lambda(z^2-1)} \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$= e^{-\lambda} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-(1-2\lambda)z^2/2} dz$$

$$\begin{aligned}
&= e^{-\lambda} \int_{-\infty}^{+\infty} \frac{\sigma'}{\sqrt{2\pi} \sigma'} e^{-\frac{z^2}{2} \underbrace{\left(\frac{1}{\sqrt{1-2\lambda}}\right)^2}_{\sigma'}} dz \\
&= e^{-\lambda} \underbrace{\sigma'}_{1} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi} \sigma'} e^{-\frac{z^2}{2\sigma^2}} dz \\
&= \frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{\frac{2\lambda^2}{2}} \quad 1-2\lambda > 0
\end{aligned}$$

$$(d=2, \alpha=4)$$

Claim:

$$\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{2\lambda^2}, \quad |\lambda| \leq \frac{1}{4}$$

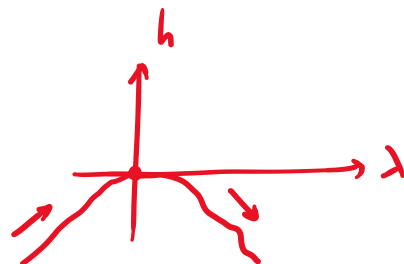
take log

$$-\lambda - \frac{1}{2} \log(1-2\lambda) \stackrel{?}{\leq} 2\lambda^2$$

$$h(\lambda) := -2\lambda^2 - \lambda - \frac{1}{2} \log(1-2\lambda) \leq 0 \quad \lambda \in [-\frac{1}{4}, \frac{1}{4}]$$

$$h(0) = 0$$

$$h'(\lambda) = -4\lambda - 1 + \frac{1}{1-2\lambda}$$



$$\begin{aligned}
&= -4\lambda + \frac{-\cancel{1} + 2\lambda + \cancel{1}}{1-2\lambda} = \lambda \left[-4 + \frac{2}{1-2\lambda} \right] \\
&= \lambda \left[\frac{-2 + 8\lambda}{1-2\lambda} \right] \stackrel{?}{=} 2(-1+4\lambda)
\end{aligned}$$

$$-\frac{1}{4} < \lambda < 0$$

$$h'(\lambda) > 0$$

$$0 < \lambda < \frac{1}{4}$$

$$h'(\lambda) < 0$$

$$\Rightarrow \mathbb{E} \left(e^{\lambda(Z^2 - 1)} \right) \leq e^{2\lambda^2} \quad \lambda \in [-\frac{1}{4}, \frac{1}{4}]$$

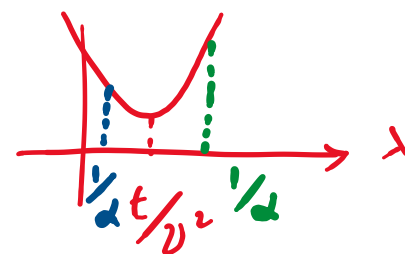
$$Z^2 \sim \text{sub E}(\nu=2, \alpha=4).$$

Goal: Find the tail prob. of $X \sim \text{sub E}(\nu, \alpha)$ $\square$

$$\begin{aligned} \mathbb{P}(X - \mu \geq t) &\stackrel{(\text{Chernoff})}{\leq} \inf_{\lambda \in [-\frac{1}{2}, \frac{1}{2}]} \mathbb{E} \left(e^{\lambda(X - \mu)} \right) \cdot e^{-\lambda t} \\ &\leq \inf_{\lambda \in [-\frac{1}{2}, \frac{1}{2}]} e^{\nu^2 \lambda^2 / 2} \cdot e^{-\lambda t} \end{aligned}$$

$$\leq \inf_{\lambda \in \dots} e^{\nu^2 \lambda^2 / 2 - \lambda t}$$

$$\nu^2 \lambda - t = 0 \Rightarrow \lambda = t / \nu^2$$



$$\text{Case 1: } t / \nu^2 \leq \frac{1}{2} \quad t \leq \frac{\nu^2}{2} \quad \mathbb{P}(X - \mu \geq t) \leq e^{-\frac{t^2}{2\nu^2}}$$

$$\text{Case 2: } t / \nu^2 \geq \frac{1}{2} \quad \lambda^* = \frac{1}{2}$$

$$\mathbb{P}(X - \mu \geq t) \leq e^{\nu^2 / 2 \alpha^2 - \frac{t}{\alpha}}$$

$$\begin{aligned}
 & \frac{v^2}{\alpha} \leq t \\
 & \leq e^{\frac{t}{2\alpha} - \frac{t}{\alpha}} \\
 & = e^{-\frac{t}{2\alpha}} \\
 P(X - \mu \geq t) & \leq \begin{cases} e^{-\frac{t^2}{2v^2}} & t \leq \frac{v^2}{\alpha} \\ e^{-\frac{t}{2\alpha}} & t > \frac{v^2}{\alpha} \end{cases}
 \end{aligned}$$

Exercise 2.3.

$X \geq 0$.

$E(X) = \dots$

$$E(e^{\lambda(X - \mu)}) < \infty$$

$$\lambda \in [-\beta, \beta].$$

$$\inf_{k=0,1,\dots} \frac{E\{|X|^k\}}{s^k} \leq \inf_{\lambda > 0} \frac{E\{e^{\lambda X}\}}{e^{\lambda s}}.$$